

Course: B. Tech.

Semester: VII

Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/

Electrical & Electronics Engg. / Electrical & Power Engineering

Subject Code & Name: Power System Operation and Control (BTEEC702)

Max Marks: 60

Duration: 3 Hr.

Date: 03/07/2024

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

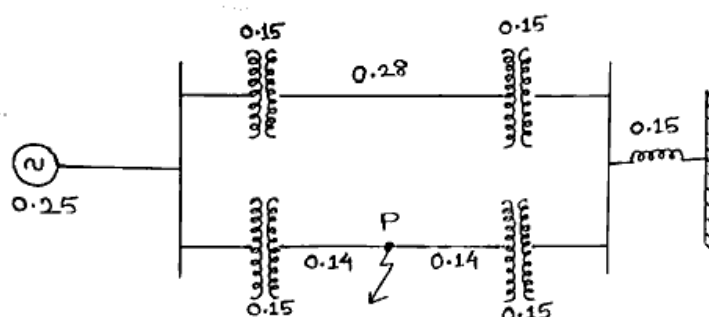
12

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|--|-----------|----------|
| A) Explain production and absorption of reactive power by different power system equipments. | L2 | 6 |
| B) Explain the application of Shunt Reactor for voltage control with the help of phasor diagram. | L2 | 6 |
| C) Compare Shunt Capacitor with Series Capacitor. | L2 | 6 |

Q.2 Solve Any Two of the following.

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| A) Classify power system stability. Explain in detail. | L2 | 6 |
| B) The figure below shows a generator connected to an infinite busbar through a double circuit transmission line. Per unit reactances are indicated. Compute critical clearing angle for a three phase fault at point P when the generator is delivering 1.0 pu power. Generator voltage is 1.2 pu and infinite busbar voltage is 1.0 pu. | L4 | 6 |



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| C) Explain point by point solution of swing equation. | L2 | 6 |
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Q.3 Solve Any Two of the following.

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|----|--|----|---|
| A) | Explain the construction and working of Brushless excitation system with neat sketch. | L2 | 6 |
| B) | What control and protective functions does an excitation system perform? Explain with block diagram. | L2 | 6 |
| C) | Explain the construction and working of Static excitation system with neat sketch. | L2 | 6 |

Q.4 Solve Any Two of the following. 12

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|----|--|----|---|
| A) | Explain Load Frequency Control of single area system. | L2 | 6 |
| B) | Explain any two of the following load frequency control methods: <ul style="list-style-type: none"> • Flat frequency control • Parallel frequency control • Flat Tie Line Control | L2 | 6 |

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|----|--|----|---|
| C) | The B matrix for a two plant system is given by: | L3 | 6 |
|----|--|----|---|

$$B = \begin{bmatrix} 0.001 & -0.0001 \\ -0.0001 & 0.0013 \end{bmatrix} MW^{-1}$$

The economic schedule for a certain load is given as:

$P_1 = 150$ MW and $P_2 = 275$ MW.

Calculate the penalty factor of plant 1.

Q.5 Solve Any Two of the following. 12

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|----|--|----|---|
| A) | Derive transmission line coefficients for a system of two power plants supplying a common load. | L2 | 6 |
| B) | Explain the objectives, constraints and solution methods associated with Hydrothermal Scheduling | L2 | 6 |
| C) | The fuel input per hour of plant 1 and plant 2 is given as: | L4 | 6 |

$$F_1 = 0.2P_1^2 + 40P_1 + 120 \text{ rupees per hour}$$

$$F_2 = 0.25P_2^2 + 30P_2 + 150 \text{ rupees per hour}$$

Determine the economic operating schedule and the corresponding cost of generation if the minimum and maximum loading on each unit is 25MW and 100 MW respectively. Total demand is 180 MW and transmission losses are neglected. If the load is equally shared by both the units, determine the saving obtained by loading the units as per equal incremental fuel cost.

*** End ***